

Responses to the blue annotations in Final/Last/main-new.tex

May 15, 2026

Each entry below quotes the blue comment, gives a concrete suggestion of what to insert (or how to rewrite), and explains the rationale. Line numbers refer to `main-new.tex` as of the current draft.

Introduction (§1.1–1.3)

L790 *Ainda não falaste de integrabilidade. Talvez pôr “see below”.*

Suggested text: “... to reach this threshold by perturbative means, that is, without appealing to the integrable structure of (??) (this point is made precise in the overview of §1.2 and in Remark [ref]).”

Why: “Integrable structure” is used here for the first time as if already discussed. A forward pointer turns an apparent gap into a deliberate signpost; alternatively, drop the clause here and introduce integrability only in §1.2 where the perturbative/non-integrable contrast is actually developed.

L834 *Uniform pode ser enganador, o tempo de existência depende de E não-uniformemente.*

Suggested text: Replace “and the statement is uniform in the energy $E \in \mathbb{R}$ ” by “and the statement holds for every fixed energy $E \in \mathbb{R}$ (the proof, and in particular the local existence time, depending on E).”

Why: “Uniform in E ” wrongly suggests E -independent constants/lifespan. The genuine uniformity is structural (same mechanism for all E), not quantitative; the rephrasing keeps the intended message without overclaiming.

L849 / L1265 *Nas transformadas de Fourier aparece muitas vezes $\mathcal{F}_z$. Dois problemas: não se fez a identificação $z = x + iy$ e z é usado para a gauged variable. Talvez pôr $\mathcal{F}_{x,y}$ ou $\hat{\cdot}$. (L1265: trocar notação de $\mathcal{F}$.)*

Suggested text: Globally replace $\mathcal{F}_z$ and $\mathcal{F}_x$ by a single symbol, e.g. $\hat{\cdot} = \mathcal{F}_{x,y}$, the spatial Fourier transform on $\mathbb{R}_{x,y}^2$; reserve z exclusively for the gauged variable.

Why: z is overloaded: as the complex coordinate $x + iy$ (never explicitly identified) and as the gauged unknown. A neutral spatial-Fourier notation removes the clash and the missing $z = x + iy$ identification altogether. This must be applied consistently (hence the same fix resolves L1265).

L864 *Lá à frente falas de absorver as constantes multiplicativas para dentro do K como discutido aqui, mas aqui não há discussão nenhuma.*

Suggested text: Insert after (??): “Here and below, absolute multiplicative constants arising from the Fourier normalisation and from the algebra of (??) are absorbed once and for all into the bilinear multiplier K ; this convention is what is referred to as ‘the discussion preceding (??)’.”

Why: Line 1230 explicitly says “see the discussion preceding (??)”, but no such discussion exists. One sentence makes the later back-reference correct.

L873 (a) *Não percebi. Talvez “summing up the boundary terms in the normal-form expansion”?*

Suggested text: “... the gauge transform $\mathcal{G}_\delta$, defined by summing the *boundary* terms of the infinite normal-form expansion.”

Why: The gauge is built precisely from the series of boundary contributions (cf. L1636–1643). “Summing up the normal-form expansion” is ambiguous and could be read as summing every term; the reviewer’s reading is the correct one.

L873 (b) *Pôr como tag?* (about the gauged NV equation).

Suggested text: Give the displayed equation a permanent tag, e.g. a `\tag{gNV}` (or a numbered `equation` environment with `\label{z1}`), and refer to it as “the gauged Novikov–Veselov equation (gNV)” throughout.

Why: The equation is cited repeatedly (“(??)"); a stable, mnemonic tag improves readability and guarantees the reference resolves.

L887 *Trocar b por $1/2^+$ ou então dizer $b = \frac{1}{2}^+$.*

Suggested text: Append to the displayed estimate “... , with $b = \frac{1}{2}^+$ and $0+$ as in the notation of §1.3.”

Why: b has not been fixed at this point in the overview, so the bound is not self-contained. Pinning $b = \frac{1}{2}^+$ (or substituting the value directly) removes the ambiguity.

L888 *it ensures summability over* (replacing “it is summable over”).

Suggested text: Accept: “... the exponent $(2s + 2)(j - 3) - 1$: it ensures summability over j for any $s > -1$...”

Why: An exponent is not itself “summable”; it is the series $\sum_j C^j \delta^{(2s+2)(j-3)-1}$ that converges. The edit is a precise and costless improvement.

L893 *Aqui não se percebe de onde vem este n .* (“uniform-in- n control”).

Suggested text: Either introduce the smoothing sequence first — “approximating the datum by a sequence $u_{0,n} \in H^\infty$ and writing u_n for the corresponding smooth solutions” — or replace “uniform-in- n ” by “uniform along the smoothing sequence”.

Why: n appears with no antecedent; it is the index of the smoothing approximation used in the density argument. Naming the sequence before using n makes the sentence self-contained.

L897 *O que se ganha não é a analiticidade (isso já está no nosso teorema), mas sim o facto de a solução viver em $X^{s,b}$.*

Suggested text: Retitle the appendix: “**Appendix: persistence in $X^{s,b}$ for $s > -\frac{3}{4}$** ”, and rewrite the body so the stated gain is the $X^{s,b}$ contraction/persistence of the flow (analyticity being already part of Theorem ??).

Why: Theorem ?? already yields analyticity in its range, so advertising “analyticity” as the appendix’s contribution is misleading. The real added value is the stronger functional setting $(X^{s,b})$, which should headline the appendix.

L898 (a) *Só era conhecido no caso $E = 0$, talvez reescrever.* (“the (already-known) range $s > -\frac{3}{4}$ ”).

Suggested text: “A short appendix is devoted to the range $s > -\frac{3}{4}$, known for $E = 0$ by Adams–Grünrock [?] and here extended to all $E \in \mathbb{R}$.”

Why: The prior result is for $E = 0$ only; “already-known” overstates what was available and hides the actual novelty of the appendix (the $E \neq 0$ case).

L898 (b) *corrigir* (“... from continuous to analytic in that range”).

Suggested text: In line with L897: “... and shows that in this range the data-to-solution map is, in addition, real-analytic and the solution persists in $X^{s,b}$.”

Why: Must be reconciled with the L897 correction: state the genuine improvement ($X^{s,b}$ persistence; analyticity recovered by an easier argument) rather than a continuous-to-analytic “upgrade” that the main theorem already provides.

L909 (a) *or $\cdot^\vee$* (alternative inverse-Fourier notation).

Suggested text: Accept: “... by $\mathcal{F}$ and $\mathcal{F}^{-1}$ (or $\hat{\cdot}$ and $\cdot^\vee$) the spatial Fourier transform and its inverse...”

Why: Introducing $\cdot^\vee$ here legitimises its use later and pairs naturally with $\hat{\cdot}$; harmless and improves consistency.

L909 (b) *defined in Section 2.2* (about $\mathcal{N}_\ell(\mathbf{T}; \cdot)$).

Suggested text: Accept: “... the multilinear operator associated with a tree $\mathbf{T}$ and a type label $\ell \in \{1, \dots, 5\}$, defined in Section ??” (use the actual label of §2.2).

Why: $\mathcal{N}_\ell$ is non-trivial; a forward reference to its definition is expected in a notation list.

Normal-form derivation (§2)

L965 *, given $\mathbf{T} \in \mathcal{T}_j$,*

Suggested text: Accept the insertion: “From Definition ??, it follows that, given $\mathbf{T} \in \mathcal{T}_j$, ...”

Why: The clause that follows quantifies over a tree of the j -th generation; naming it explicitly removes a dangling reference.

L1214 *Isto não está coerente. Γ é um conjunto em $\mathbb{R}^3$, A em $\mathbb{R}^2$...*

Suggested text: Make the ambient space uniform. Either present both as subsets of the convolution hyperplane,

$$A := \left\{ (\xi_1, \xi_2) \in (\mathbb{R}^2)^2 : \xi_1 + \xi_2 = \xi, |\xi_1| \vee |\xi_2| \leq 1/\delta \right\}, \quad A^c := \Gamma([1]) \setminus A,$$

defining $\Gamma([1])$ as that same hyperplane $\{\xi_1 + \xi_2 = \xi\}$; or carry the constraint $\xi_1 + \xi_2 = \xi$ explicitly so that $A \subset \Gamma([1])$ literally.

Why: As written, $\Gamma([1]) \setminus A$ subtracts an $\mathbb{R}^2$ -set from an $\mathbb{R}^3$ -set, which is ill-typed. Fixing $\Gamma([1])$ and A to live on the same constraint set makes $A^c = \Gamma([1]) \setminus A$ meaningful.

L1525 *Referir que a prova está feita em [11] (é preciso explicar o que acontece com os termos E ?).*

Suggested text: Add: “The systematic iteration is carried out exactly as in [?]; the only new feature here is the energy term: the splitting $K\Psi_0 = K\Psi - EK\Psi_E$ feeds the lower-order piece $EK\Psi_E$ into the remainders, where it is estimated perturbatively (see (??)), so the combinatorics of the trees is unaffected.”

Why: The reader needs to know the iterative machinery is borrowed (cite the source) and that the E -dependent terms do not alter the tree bookkeeping — the natural worry given (??) carries the extra $E\Phi_E$ symbol.

L1643 *Não faleste the colourings.*

Suggested text: Insert before first use: “Each leaf of a tree may be filled either with w or with $\mathcal{D}[w, w]$; we call such an assignment a *colouring* of the tree. Expanding (??) thus produces, for each underlying tree, several terms of equal homogeneity indexed by their colouring.”

Why: “Colourings” is introduced as if defined; one sentence makes the combinatorial object precise and the “bundling tree-by-tree” statement rigorous.

L1645 *Pôr isto como título do lemma?* (the heading “The boundary cancellation”).

Suggested text: Drop the \subsubsection* and fold the phrase into the lemma, e.g. “**Lemma ??** (Boundary cancellation).”.

Why: The heading and Lemma ?? state the same fact; naming the lemma “Boundary cancellation” avoids a redundant sectioning unit and gives the result a citable name.

Multilinear estimates (§3)

L1873 *O Lema ?? não é de frequência restrita, eu tiraria este.*

Suggested text: “... funnelled into one of two off-the-shelf frequency-restricted estimates [?, ?] (Lemmas ??, ??), combined with the elementary integration bound Lemma ??...”

Why: Lemma ?? is an auxiliary $\langle \cdot \rangle$ -integration estimate, not a frequency-restricted estimate; listing it among the FRE inputs mischaracterises it. Demoting it to a separate “combined with” clause is accurate and keeps it in the argument.

L2011 *Passar a referências?* (Hörmander / Duistermaat cited inline).

Suggested text: Add `hormander1`, `duistermaat` to `biblio.bib` and replace the inline mentions by “Morse’s lemma with parameters ([?, Lemma C.6.1]; see also [?, §3.4])”.

Why: Books cited inline by title break the citation style used everywhere else; bib entries make them consistent and clickable.

L2073 *★1*, inserted into the heading “ $B = \{★1, ★2\}$ ”.

Suggested text: Reconcile heading and body: the text below says “Choosing $B = \{★2\}$ ”. Decide which is correct and make both read identically (most likely $B = \{★2\}$, since the argument trades resonance only against $\xi_{★2}$).

Why: The inserted “★1,” contradicts the body of Step 2; an inconsistency in the choice of the splitting set B would confuse the reader following the bookkeeping. Verify against the multiplier and unify.

L2074 *large* (“via the large-frequency restriction $|\xi_{★2}| \gtrsim 1/\delta$ ”).

Suggested text: Accept: “... contributes a factor δ^{2s+2} via the large-frequency restriction $|\xi_{★2}| \gtrsim 1/\delta$...”

Why: $|\xi_{★2}| \gtrsim 1/\delta$ is precisely a high/large-frequency constraint; the qualifier removes ambiguity about which regime supplies the δ^{2s+2} gain.

L2092 *2*, inserted into “ $B = \{2, 11\}$ ”.

Suggested text: Accept provided it matches the body of Step 3: with $\mathbf{T} = [21]$ and the worst-case ordering $|\xi_2| \geq |\xi| \geq |\xi_{11}| \geq |\xi_{12}|$, the splitting set should indeed be $B = \{2, 11\}$. Confirm against (??) and the integration that follows, then keep $B = \{2, 11\}$.

Why: The choice of B must coincide with the variables actually traded against resonance in Step 3; the insertion looks correct but should be checked against the displayed multiplier before being made permanent.

L2265 *Isto não está certo assim.* (“ $\int \langle \xi_{p(★)} \rangle^{-(4s+2)} d\xi_{p(★)} \lesssim 1$ for $s > -1$ ”).

Suggested text: Correct the convergence claim. Since $\xi_{p(★)} \in \mathbb{R}^2$, the plain integral $\int_{\mathbb{R}^2} \langle \xi \rangle^{-(4s+2)} d\xi$ converges only for $4s + 2 > 2$, i.e. $s > 0$, *not* for $s > -1$. The fix is to keep the resonance restriction or an extra weight on $\xi_{p(★)}$ so the effective exponent exceeds the dimension 2: e.g. retain a factor $\mathbf{1}_{|\Psi(\mathbf{T})-\alpha|<M}$ (or the companion $\langle \xi_{★21} \rangle$ weight) in the $\xi_{p(★)}$ integration and conclude $\int \langle \xi_{p(★)} \rangle^{-(4s+2)} \mathbf{1}_{\{\cdot\}} d\xi_{p(★)} \lesssim M$ (or $\lesssim 1$) by Lemma ??, rather than asserting unrestricted convergence for $s > -1$.

Why: In two dimensions $\langle \xi \rangle^{-(4s+2)}$ is integrable iff $s > 0$, so the sentence as written is false in the relevant range $s \in (-1, 0]$. The gain claimed for Step 7 must come from the resonance/weight structure, not from bare integrability; the surrounding $\delta^{(2s+2)(j-3)}M$ conclusion is likely still correct but the intermediate justification needs to be redone with the restriction kept.

Inversion / density (§4)

L2540 δ inserted into “ $\sum_{j \geq 0} (C\delta)^j$ ”.

Suggested text: Accept: the series must read $\sum_{j \geq 0} (C\delta)^j$ (geometric, convergent once $C\delta < 1$).

Why: Without the δ the series $\sum_j C^j$ diverges; the smallness that closes the contraction/uniqueness estimate is exactly the gauge parameter δ (taken small in terms of the data), so $(C\delta)^j$ is the intended summand. State explicitly that δ is chosen with $C\delta < 1$.

Cross-cutting note. Items L849, L909(a), L1265 are the same underlying issue (Fourier-transform notation and the symbol z); fixing them together with a single global convention is preferable to local patches. Items L897, L898(a), L898(b) must also be edited together so the appendix’s stated contribution is internally consistent.